

# Daily CCS Exec Briefing

For Ayu · Geologist, PETRONAS CCS Track

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Storage Subsurface Intelligence · Pattern-level Analysis

Compiled by Hermes · arifOS Federation

⚠ **Source caveat:** Live news tickers not reachable (SearXNG down, DuckDuckGo timeout, Google blocked). This briefing combines locked memory intel with confirmed signal patterns — not fresh tickers. Items flagged "verify" require Ayu's own cross-check via Reuters Energy / Upstream / Energy Voice.

# 1. PETRONAS Kasawari CCS — Status Track

STORAGE CAPACITY TARGET

~3.3 Mtpa CO<sub>2</sub>

## Project Anchors

- **FID announced 2023** — committed investment decision locked
- **Reservoir:** depleted M4 carbonate, offshore Sarawak
- **Injection window:** public expectation 2025–2026 (exact start date not locked in memory — Ayu knows operational reality better than this brief)

## Risk Surface (Top 3)

### Caprock Fault Reactivation

CO<sub>2</sub>-induced pressure buildup reopens existing faults under caprock → seal breach pathway

### Legacy Well Leakage

Old wells through caprock = pre-existing leakage pathways, P&A status critical

### Plume Migration Uncertainty

Reactive transport simulation gap — CO<sub>2</sub>-brine-rock mineral trapping in limestone vs sandstone

## Monitoring Signals to Watch

Well integrity test reports

Microseismic monitoring data

4D time-lapse seismic (plume tracking)

Halokinesis effect on M4 caprock

Source: memory intel + Kasawari public disclosures → verify with internal PETRONAS project tracker

## 2. Malaysia CCS Regulatory — CCUS Bill

### PSL Framework — Unresolved

Post-storage liability transfer (30+ years post-closure): operator → state — **who holds it?**

### Three Critical Regulatory Questions

| Question                       | Current Status                         | Ayu-Impact                                                         |
|--------------------------------|----------------------------------------|--------------------------------------------------------------------|
| Long-term liability transfer   | <b>Drafting</b>                        | Permit applications must pre-allocate liability pathway            |
| MRV compliance framework       | <b>In development</b>                  | Reporting standards still fluid — pre-empt with IPCC-tier baseline |
| P&A standards for legacy wells | <b>Refer to offshore oil &amp; gas</b> | CCS-specific requirements being drafted — watch this space         |

### ⚠ Cross-check required

Actual CCUS Bill Parliament status (2nd reading? passed?) — confirmation tickers not reachable today

# 3. Southeast Asia CCS Geopolitics

## Indonesia

### ADVANCING

Aggressive CCS-as-service offering, attracting Western majors. Positioning as regional CCS hub with depleted field inventory across Sumatra & East Kalimantan.

## Singapore

### STUDY PHASE

Cross-border CO<sub>2</sub> transport chain to Malaysia/Indonesia storage sites still in feasibility stage. Hub for CO<sub>2</sub> aggregation from regional emitters.

## Malaysia / Sarawak

### STRATEGIC ASSET

Kasawari-class storage in depleted offshore fields. Deepwater proximity to high-emission East Asia. Existing oil & gas infrastructure reduces capex baseline.

## Japan / Korea

### DEMAND SIDE

Off-takers driving blue hydrogen + CCS chain. Sarawak H<sub>2</sub> hub depends on Kasawari-class storage for compliance with Japan's hydrogen import standards.

### STRATEGIC IMPLICATION FOR PETRONAS

Sarawak storage advantage is real and time-limited. Competitor entry (Indonesia) = pricing pressure downstream. First-mover position on operational storage = asset value capture before market normalization.

## 4. Geoscience Technical Pulse — What CCS Geologists Must Watch

01

### Injectivity vs Capacity

Pressure buildup modeling & fracture gradient. M4 carbonate porosity/permeability baseline.

02

### Caprock Integrity

Seal capacity under re-pressurization. Shale gouge ratio, capillary entry pressure thresholds.

03

### Plume Migration

Reactive transport simulation (CO<sub>2</sub>-brine-rock). Mineral trapping potential limestone vs sandstone.

04

### Legacy Well Integrity

P&A status, cement bond logs, sustained casing pressure tests. Old wells = leak pathways.

05

### Monitoring Protocol

InSAR, 4D seismic, microseismic, fluid sampling. Frequency dictated by regulator — still evolving.

06

### MRV Framework

IPCC-tier accounting. Malaysia needs MOC (margin of conservatism) framework development.

**Global benchmarks to study:** Quest CCS (Canada) · Gorgon (Australia) · Northern Lights (Norway) · Sleipner (Norway) — published post-mortems available, all include operational lessons learned.

# Bottom Line — What Ayu Should Do

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## IF HANDLING SUBSURFACE WORKSTREAM

Your focus right now is **caprock integrity modeling + legacy well audit**. These two are the global industry bottleneck — operational excellence here is competitive advantage.

## IF HANDLING STAKEHOLDER COMMUNICATION

Be ready — Malaysia CCS regulatory uncertainty is still real. **Every project is bespoke negotiation** until the CCUS Bill passes and PSL framework is settled.

## IF NEW TO CCS

Monitor global incidents (Quest CCS Canada, Gorgon Australia, Northern Lights Norway) — their post-mortem published reports are all available. **That is your core curriculum.**

## CROSS-CHECK REQUIRED TODAY

Kasawari injection start date · CCUS Bill Parliament status · Bursa Carbon Exchange spot price · any PETRONAS press release this morning.

**Search backends down — verify via Reuters / Upstream / Energy Voice.**